JÖNKÖPING UNIVERSITY
School of Engineering

Cyber Terrorism and Information Warfare

Joakim Kävrestad

TODAY

- Cyber terrorism and cyber warfare
  - What is it
  - Some examples
- Information warfare
  - What is it
  - Some examples
- What can we do about it?

CLASSIFYING CRIME

- ”Private” crimes
  - Fraud, drugs, stalking, sex offenses etc
- Crimes against companies
  - Industrial espionage
- **Crimes against states**
  - **Cyber warfare**

ALL THOSE TERMS ☹

- Cyber terrorism
  - is the premeditated, politically motivated attack against information, computer systems, computer programs, and data that results in violence against **noncombatant targets** by subnational groups or clandestine agents.
- Cyber warfare (not defined in the book)
  - Is **interstate** use of cyberspace with the intent to threaten a nations security or a response to a perceived threat against a nations security
- Information warfare
  - is any attempt to manipulate information in pursuit of a military or political goal.

CYBER TERRORISM AND WARFARE

- Terms often used interchangeably
  - Terrorism is arguably broader
- Quite common to see some level of hybrid warfare
  - Cyber elements in traditional warfare
- To some extent, cyber warfare and information warfare seems to be common in times or peace

CYBER TERRORISM AND WARFARE

- The general idea is to user cyberspace to reach military and/or political goals

- Those goals can be very different so lets consider a few threat agents
  - Russia (vs Ukraine)
  - Russie (vs Sweden)
  - USA (vs Germany)
  - Anonymous

CYBER TERRORISM AND WARFARE

- Aiding warfare
  - Attacking hospitals
  - Attacking electric systems
  - Attacking communications systems

- Deterrence
  - Disrupt service
  - Scare tactics

CYBER TERRORISM AND WARFARE

- Ensuring compliance?
  - Spy on your friends
  - Keeping allies in line?

- Political statements
  - Taking organizations you don’t like offline
  - Scaring policymakers into one or another decition

Reflect:1-2-4-all

Individually (60 Sec): Write core statements (one-liners) about what you learned in the last lecture part.

Pair up and compare lists (60 Sec) (add new things to your individual lists)

Pair the pairs and repeat (60 Sec)

Then one statement per group of four.

STUXNET 1

- Stuxnet is sometimes called the first cyber warfare act
  - Worm which targeted Iranian nuclear program
  - Used to destroy uranium enrichment centrifuges in Iranian power plants (or something, no questions on nuclear stuff please☺)
- Supposedly done by USA and Israel
- Based on a worm activated at a certain time

STUXNET 2

- Infection
  - Via an inserted USB drive
  - Supposedly a road apple

- Spread
  - Uses various vulnerabilities to spread across a network
  - If a computer have a specific siemens software for controlling those centrigues, it infects that. Otherwise it does nothing

STUXNET 3

- Action
  - At a given event or time, it will start meddling with the control software.
  - Eventually destroys the centrifuges
  - Also ensures that “reasonable values” are displayed by the control software

- Discovery
  - Spreads like the wind
  - Discovered in millions of computers…even if it does no harm to most of those

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Break!

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INFORMATION WARFARE

- Fake news
- Promoting your truth
- Controlling information obtained by your opponents

FAKE NEWS/DISINFORMATION

- Promoting and propagating fake or sewed messages

- The idea is to build a narrative to reach some objectives
  - Confuse
  - Harm
  - Create instability
  - Impact on policymaking and similar

SWEDEN STEALS YOUR CHILDREN!

- A campaign promoting the idea that Swedish government would steal (predominantly Muslim) children
- Led to protests and bojkots agains Sweden and Swedish interests
- May have made the NATO application more difficult
- Supposedly started by Islamist extremists

https://www.fhs.se/download/18.32d29dd2187bd01d5e455265/1682576119173/LVU-kampanjen.pdf

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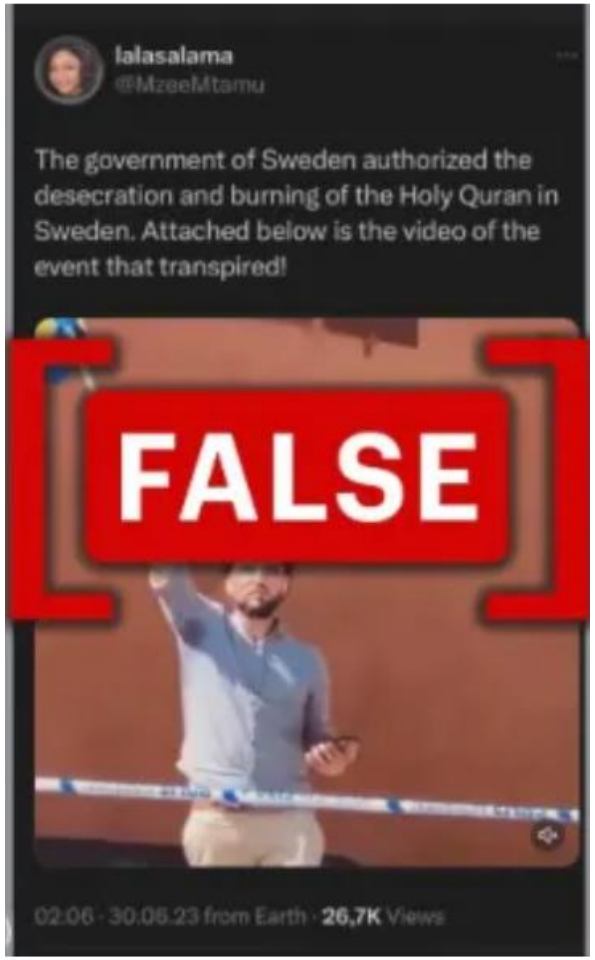

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PROMOTING YOUR TRUTH

- In many cases, there are two sides or a coin but no absolute truth

- Examples
  - Policy decisions - we know the facts but what do we decide to do
  - Warfare – An attack at a hospital where terrorists are hiding is both an attack at hospitals and an attack on terrorists

- By controlling the narrative, you can control global politics and policymaking

EXAMPLE - BREXIT

- Brexit – the UK vote to leave EU
- Supposedly, Russian initiatives supported spreading news which supported the “Brexit-side”
- Modus examples
  - +150 000 twitter accounts with Russian tweted in support for the “Brexit” side in the run-up to the vote

EXAMPLE - COVID-19 VACCINE

- There has been plenty of different …ideas….around the covid vaccinations
- Depending on how you frame a message, the same fact can promote different points.
- Lets see how we can play with this!

EXAMPLE – COVID-19 VACCINE

- Severe reactions after COVID-19 vaccination are rare
- Anaphylaxis is a very rare side effect, which can be affectively treated
- Myocarditis is a very rare, but treatable, reaction to COVID-19 vaccination!

- Covid-19 vaccination can cause severe side effects
- Some people get anaphylaxis, which can kill you.
- Some people get myocarditis and die.

From https://www.cdc.gov/coronavirus/2019-ncov/vaccines/safety/adverse-events.html

CONTROLLING WHAT OTHERS KNOW

- A different type of information war is about obtaining information about your adversaries, and controlling what your adversaries know about you
  - You want to have information about your adversaries
  - You may also want to give those adversaries false information about yourself

- Naturally, states are spying on each other and so on

EXAMPLE- RADIO FREE EUROPE

- During the cold war, western countries spend time and money to broadcast radio in communist states
- The idea wat to impact public opinion in order to discredit communism

EXAMPLE- WW2

- During world war 2, Germany was aided by Enigma which allowed them to send secret messages
- The allied managed to crack it, which allowed the to read (some) of Germanys encrypted information
- By having access to information Germany thought was secret, the allied could know about Germanys plans ahead of time

EXAMPLE- RADIO FREE EUROPE

- Placing false information to trick enemies is common
- On example, operation mincemeat, involved the British placing a dead body with false information off the coast of Spain
  - Suggested an upcoming invasion in Greece
- There was instead an invasion in Sicily…..which came a suprice for the Germans

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TRENDS

- Positive
  - Governments are taking the problem seriously with various efforts into increased national cyber resilience
  - Cyber defense is being seen as a military branch (e.g. the cyber soldier initiative in Sweden)
- Negative
  - The problem is increasing…because…we other sides are also investing in digital

WHAT TO DO ABOUT IT

- Nation state and terrorists as adversaries
- Defending against fake news
- Ensuring that we know that information is real

WHAT TO DO ABOUT IT

- Cyber skills with law enforcement
- Contingency planning – we need to work, even when the systems are down
- Information literacy – the way we see, and process information is completely changed, and we need to adapt

Questions?

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